GCSE CHEMISTRY

Higher Tier – Paper 1 style

Topic 4.5: Energy changes

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 130 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **123**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 123

0 1 This question is about energy changes in chemical reactions.

0 1 . 1 Which statement describes an exothermic reaction? **[1 mark]**

*Tick **one** box.*

- ☐ Energy is transferred from the surroundings
- ☐ Energy is transferred to the surroundings
- ☐ No energy is transferred
- ☐ The temperature of the surroundings falls

0 1 . 2 What is meant by an **endothermic** reaction? **[2 marks]**

0 1 . 3 Tick (✓) one box in each row to show whether the change is exothermic or endothermic. **[5 marks]**

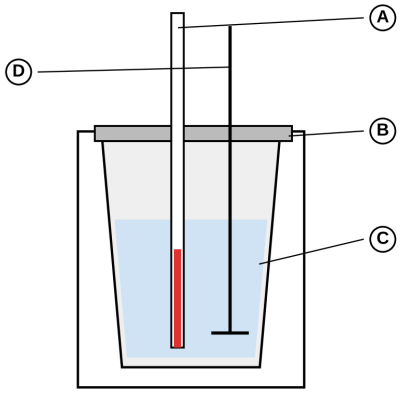
Change	Exothermic	Endothermic
burning methane		
thermal decomposition of calcium carbonate		
neutralisation of an acid by an alkali		
citric acid reacting with sodium hydrogencarbonate		
respiration		

0 1 . 4 A hand warmer contains iron powder. When the packet is opened, the iron reacts with oxygen from the air and the hand warmer gets hot. **[2 marks]**
Explain why the hand warmer gets hot.

0 1 . 5 A sports injury pack is used to treat bruises. When the pack is squeezed, two chemicals mix and the pack gets very cold.
Explain why the pack gets cold. **[2 marks]**

0 1 . 6 In an exothermic reaction the energy given out is not lost. Explain what happens to the energy. **[2 marks]**

0 2 A student investigates the temperature change when different metals are added to copper(II) sulfate solution. She uses the apparatus shown.



0 2 . 1 Name the parts labelled **A** and **B**. **[2 marks]**

0 2 . 2 Why does the student use a polystyrene cup? **[1 mark]**

0 2 . 3 Why does the student put a lid on the cup? **[1 mark]**

0 2 . 4 The student stirs the solution during the reaction. Suggest why. **[1 mark]**

0 2 . 5 The student adds 1.0 g of each metal powder to 25 cm^3 of copper(II) sulfate solution. Complete the table by calculating the temperature changes. **[2 marks]**

Metal	**Starting temperature in °C**	**Highest temperature in °C**	**Temperature change in °C**
magnesium	21.0	54.5	
zinc	21.0	41.0	
iron	21.0	26.0	
copper	21.0	21.0	

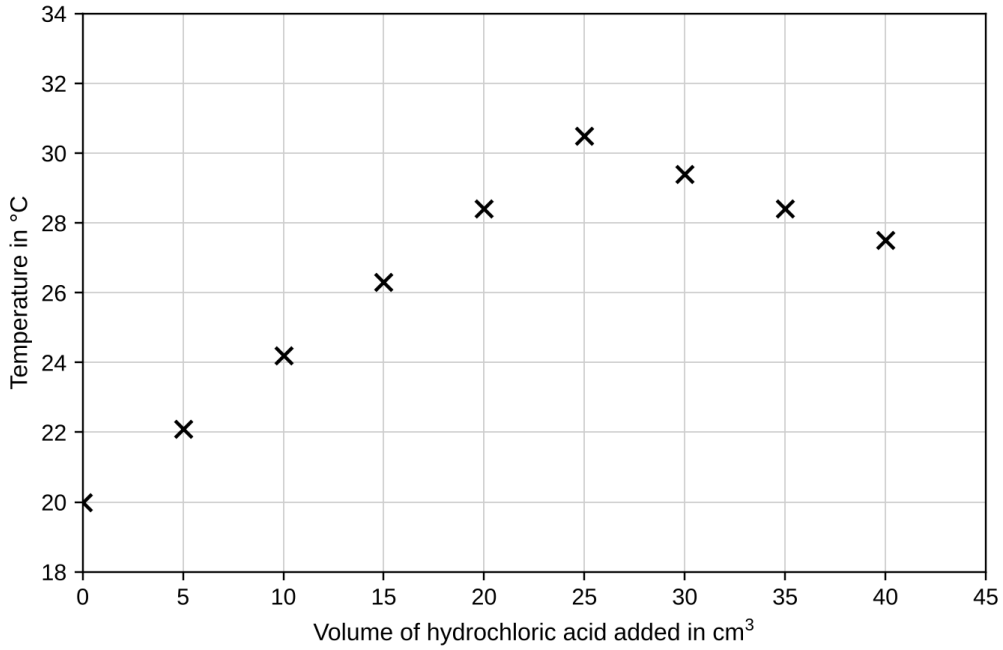
0 2 . 6 What conclusion can the student make about the reactivity of the metals and the temperature change? **[2 marks]**

0 2 . 7 Give **two** variables the student must control to make this a fair test. **[2 marks]**

0 2 . 8 How do the results show that the reaction of magnesium with copper(II) sulfate is exothermic? **[1 mark]**

0 2 . 9 Suggest **one** way the student could make the results more reliable. **[1 mark]**

0 3 A student adds dilute hydrochloric acid to 25 cm^3 of sodium hydroxide solution in a polystyrene cup. She measures the temperature after each 5 cm^3 of acid is added. The acid and the alkali have the same concentration. The student plots her results.



0 3 . 1 Draw **two** straight lines of best fit on the graph: one through the first six points and one through the last four points. Extend the lines until they cross. **[2 marks]**

0 3 . 2 Use your lines to find the maximum temperature. **[1 mark]**

0 3 . 3 Use your lines to find the volume of acid that neutralises the alkali. **[1 mark]**

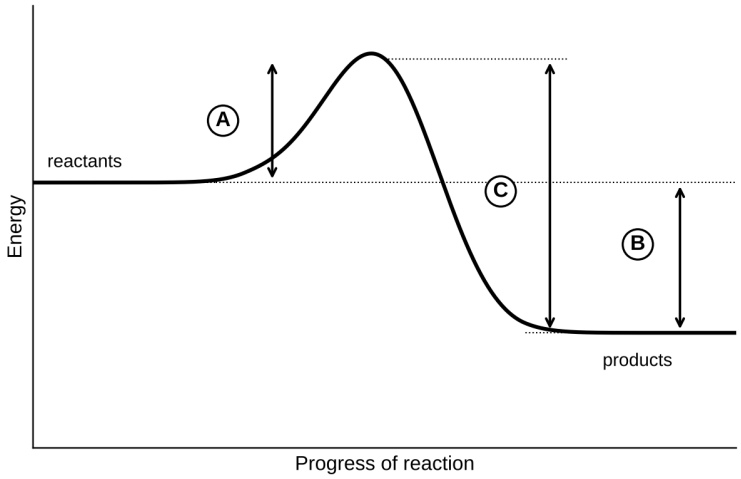
0 3 . 4 Calculate the maximum temperature rise. **[1 mark]**

Answer = _____ °C

0 3 . 5 Explain why the temperature falls after the maximum is reached. **[2 marks]**

0 3 . 6 The student repeats the experiment using sodium hydroxide solution of **twice** the concentration. The acid is unchanged. **[2 marks]**
Predict the volume of acid needed to reach the maximum temperature and what happens to the maximum temperature.

0 4 The diagram shows the reaction profile for a chemical reaction. The letters **A**, **B** and **C** show energy changes.



0 4 . 1 Which arrow shows the activation energy? **[1 mark]**

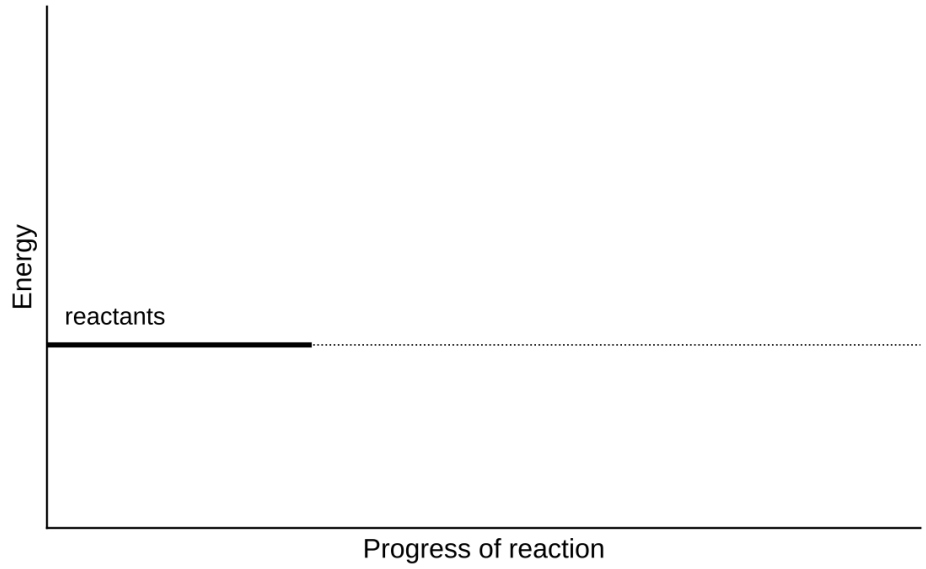
0 4 . 2 Which arrow shows the overall energy change of the reaction? **[1 mark]**

0 4 . 3 Is the reaction exothermic or endothermic? Give a reason for your answer using the diagram. **[2 marks]**

0 4 . 4 What is meant by activation energy? **[1 mark]**

0 4 . 5 Explain, in terms of bonds, why this reaction is exothermic. **[3 marks]**

0 5 The diagram shows the energy of the reactants in an **endothermic** reaction.



0 5 . 1 Complete the reaction profile for the endothermic reaction. Draw a line for the products and a curve linking the reactants to the products. **[3 marks]**

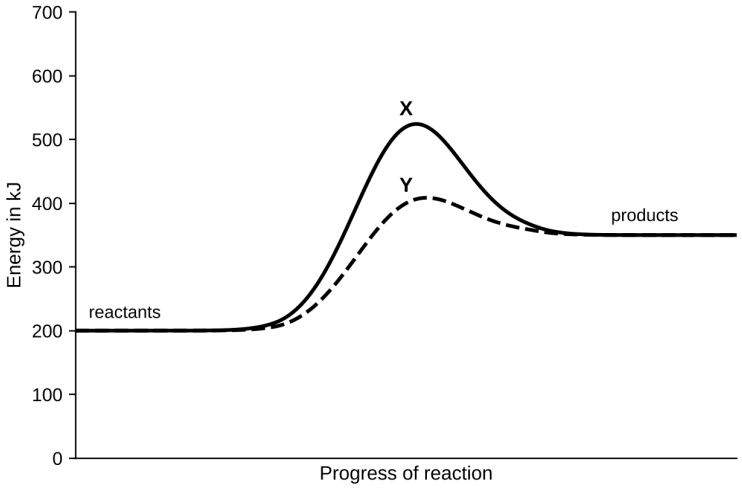
0 5 . 2 Label on your diagram (i) the activation energy and (ii) the overall energy change. **[2 marks]**

0 5 . 3 Calcium carbonate is heated strongly to make calcium oxide and carbon dioxide. The reaction is endothermic. **[2 marks]**

Explain why the calcium carbonate has to be heated continuously.

0 5 . 4 Burning methane is exothermic, but the gas will not burn until it is lit. Explain why. **[2 marks]**

0 6 The graph shows the reaction profile for a reaction with and without a catalyst. Curve **X** is for the reaction without a catalyst.



0 6 . 1 Which curve, **X** or **Y**, shows the reaction with a catalyst? Give a reason. **[2 marks]**

0 6 . 2 Is the reaction exothermic or endothermic? Give a reason. **[2 marks]**

0 6 . 3 Calculate the activation energy of the reaction **without** a catalyst. **[2 marks]**

Answer = _____ kJ

0 6 . 4 Calculate the activation energy of the reaction **with** a catalyst. **[1 mark]**

Answer = _____ kJ

0 6 . 5 Calculate the overall energy change for the reaction. Include the correct sign. **[2 marks]**

Answer = _____ kJ

0 6 . 6 Does the catalyst change the overall energy change? Give a reason. **[1 mark]**

_____

_____

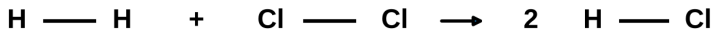
0 6 . 7 Explain how a catalyst increases the rate of a reaction. **[2 marks]**

_____

_____

_____

0 7 Hydrogen reacts with chlorine to make hydrogen chloride. The equation is shown using displayed formulae.
Bond energies are given in the table.



Bond	H–H	Cl–Cl	H–Cl
Bond energy in kJ/mol	436	242	432

0 7 . 1 Calculate the energy needed to break the bonds in the reactants. **[2 marks]**

Answer = _____ kJ/mol

0 7 . 2 Calculate the energy released when the bonds in the products form. **[2 marks]**

Answer = _____ kJ/mol

0 7 . 3 Calculate the overall energy change for the reaction. **[2 marks]**

Answer = _____ kJ/mol

0 7 . 4 Is the reaction exothermic or endothermic? Explain your answer using your calculations. **[2 marks]**

0 8 This question is about bond energies.

0 8 . 1 Methane burns in oxygen: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ **[4 marks]**

Use the bond energies in the table to calculate the overall energy change for this reaction.

Bond energies in kJ/mol: C–H 413 O=O 498 C=O 805 O–H 464

Answer = _____ kJ/mol

0 8 . 2 Ammonia is made from nitrogen and hydrogen: $N_2 + 3H_2 \rightarrow 2NH_3$ **[3 marks]**

Calculate the energy change using these bond energies in kJ/mol: N≡N 945 H–H 436 N–H 391

Answer = _____ kJ/mol

0 8 . 3 Hydrogen reacts with bromine: $H_2 + Br_2 \rightarrow 2HBr$ **[3 marks]**

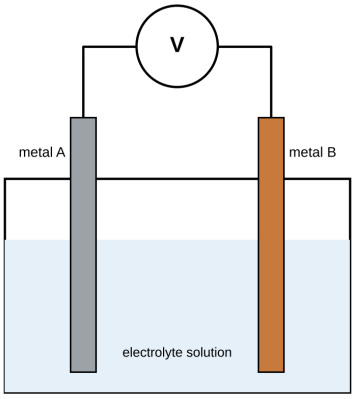
The overall energy change is −103 kJ/mol. The bond energy of H–H is 436 kJ/mol and of Br–Br is 193 kJ/mol.

Calculate the bond energy of the H–Br bond.

Answer = _____ kJ/mol

0 8 . 4 A reaction has an overall energy change of +92 kJ/mol. **[2 marks]**
What does this tell you about the energy needed to break bonds compared with the energy released when bonds form?

0 9 A student makes simple cells using copper as metal **B** and different metals as metal **A**. The electrolyte is the same each time. The student measures the voltage of each cell.



0 9 . 1 What is the name of the solution that the metals are placed in? **[1 mark]**

0 9 . 2 Which metal gives the greatest voltage? Explain why. **[2 marks]**

Metal A	Voltage in V
zinc	1.10
iron	0.78
magnesium	2.71
lead	0.47

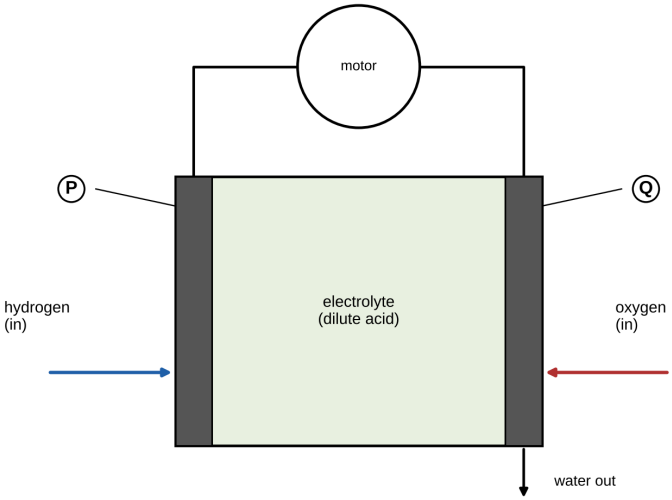
0 9 . 3 Predict the voltage of a cell made from zinc and iron with the same electrolyte. **[1 mark]**

0 9 . 4 What voltage would the student read if both metals were copper? Explain why. **[2 marks]**

0 9 . 5 An alkaline battery in a torch stops working after a few weeks. Explain why. **[2 marks]**

0 9 . 6 Explain how a rechargeable battery can be used again. **[2 marks]**

1 0 The diagram shows a hydrogen fuel cell used to power a small motor.
Hydrogen and oxygen are supplied to the cell.



1 0 . 1 Name the product of the reaction in the fuel cell. **[1 mark]**

1 0 . 2 Write the overall equation for the reaction in the fuel cell. **[2 marks]**

1 0 . 3 At which electrode, **P** or **Q**, is hydrogen oxidised? Explain your answer. **[2 marks]**

1 0 . 4 Write the half equations for the reactions at electrode **P** and electrode **Q**. **[2 marks]**
The electrolyte is acidic.

10.5 Give **two** advantages of a hydrogen fuel cell compared with a rechargeable battery. **[2 marks]**

10.6 Give **one** disadvantage of hydrogen fuel cells. **[1 mark]**

11 A student mixes citric acid solution with sodium hydrogencarbonate. The temperature falls from 21 °C to 12 °C and the mixture fizzes.

11.1 Calculate the change in temperature. **[1 mark]**

11.2 Name the gas made and describe its test. **[2 marks]**

11.3 Is the reaction exothermic or endothermic? Explain your answer. **[2 marks]**

11.4 Suggest a use for this type of reaction. **[1 mark]**

1 2 Stretch questions (grade 7–9).

1 2 . 1 *Explain, in terms of bond breaking and bond making and using reaction profiles, the difference between an exothermic reaction and an endothermic reaction. **[6 marks]**

1 2 . 2 *A company is deciding whether to use hydrogen fuel cells or rechargeable batteries to power its delivery vans. **[6 marks]**
Evaluate the use of the two types of power source.

END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																	4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.5: Energy changes

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	Energy is transferred to the surroundings	Tick the box next to 'Energy is transferred to the surroundings'.	1	AO1 4.5.1.1
01.2	a reaction that takes in energy from the surroundings ; (so) the temperature of the surroundings decreases ;	Allow 'takes in heat'. Do not accept 'needs energy' without 'from the surroundings'.	2	AO1 4.5.1.1
01.3	burning methane: exothermic ; thermal decomposition of calcium carbonate: endothermic ; neutralisation: exothermic ; citric acid + sodium hydrogencarbonate: endothermic ; respiration: exothermic ;	One mark for each correct row.	5	AO1 4.5.1.1
01.4	the reaction (oxidation of iron) is exothermic ; energy is transferred to the surroundings / hands ;		2	AO2 4.5.1.1
01.5	the reaction / dissolving is endothermic ; energy is taken in from the surroundings ;		2	AO2 4.5.1.1
01.6	energy is conserved / cannot be created or destroyed ; it is transferred from the chemicals (stored in bonds) to the surroundings ;	Allow 'as heat / thermal energy'.	2	AO2 4.5.1.1
02.1	A: thermometer ; B: lid ;	Allow 'cover'. Do not accept 'cap'.	2	AO1 4.5.1.1
02.2	it is a good insulator / reduces energy (heat) transfer to the surroundings		1	AO2 4.5.1.1
02.3	to reduce energy (heat) loss / evaporation to the surroundings		1	AO2 4.5.1.1
02.4	so the temperature is the same throughout / to mix the reactants / to get an accurate maximum temperature		1	AO3 4.5.1.1
02.5	magnesium: 33.5 ; zinc: 20.0 ; iron: 5.0 ; copper: 0.0 ; (all four correct for 2 marks; two or three correct for 1 mark) ;	Allow +33.5 etc.	2	AO2 4.5.1.1
02.6	the more reactive the metal the greater the temperature change ; magnesium > zinc > iron > copper (copper does not react) ;		2	AO3 4.5.1.1
02.7	volume of copper(II) sulfate solution ; concentration of the solution / mass of metal / particle size (surface area) of metal / starting temperature ;	Any two. Do not accept 'amount' alone.	2	AO3 4.5.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	the temperature increased / rose by 33.5 °C		1	AO2 4.5.1.1
02.9	repeat the experiment and calculate a mean ; (ignore anomalies)	Allow 'use a data logger / digital thermometer' for more accurate.	1	AO3 4.5.1.1
03.1	line through points at 0–25 cm^3 (rising) ; line through points at 25–40 cm^3 (falling), extended to meet the first line ;	Lines must be straight, ruled, with a balance of points either side.	2	AO2 WS 3.5
03.2	30.0 – 31.0 °C	Allow 30.5 from the highest point if no lines drawn.	1	AO3 4.5.1.1
03.3	24 – 26 cm^3		1	AO3 4.5.1.1
03.4	10.0 – 11.0 °C (answer from Q03.2 minus 20.0)		1	AO2 4.5.1.1
03.5	all the alkali has been neutralised (no more reaction) ; extra (cooler) acid cools the mixture / energy is lost to the surroundings ;		2	AO3 4.5.1.1
03.6	50 cm^3 ; the maximum temperature is higher ;	Twice as many moles of alkali need twice as much acid. More energy is released.	2	AO3 4.5.1.1
04.1	A		1	AO1 4.5.1.2
04.2	B		1	AO1 4.5.1.2
04.3	exothermic ; the products have less energy than the reactants / energy is released to the surroundings ;		2	AO2 4.5.1.2
04.4	the minimum energy that particles must have to react / when they collide		1	AO1 4.5.1.2
04.5	energy is needed to break the bonds in the reactants ; energy is released when new bonds form in the products ; more energy is released making bonds than is taken in breaking bonds ;		3	AO2 4.5.1.3
05.1	products level drawn above the reactants ; curve rises from the reactants to a peak ; the peak is above the products and then falls to the product level ;		3	AO2 4.5.1.2
05.2	activation energy arrow from the reactants to the top of the peak ; overall energy change arrow from the reactants level to the products level ;		2	AO2 4.5.1.2
05.3	the reaction takes in energy from the surroundings ; so energy must be supplied all the time (otherwise the reaction stops) ;		2	AO2 4.5.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	energy is needed to start the reaction / activation energy ; to break bonds in the reactants / so that collisions have enough energy ;		2	AO3 4.5.1.2
06.1	Y ; it has a lower activation energy ;		2	AO2 4.5.1.2
06.2	endothermic ; the products have more energy than the reactants (350 kJ is greater than 200 kJ) ;		2	AO2 4.5.1.2
06.3	520 − 200 ; 320 (kJ) ;	Correct answer with no working scores 2.	2	AO2 4.5.1.2
06.4	400 − 200 = 200 (kJ)		1	AO2 4.5.1.2
06.5	350 − 200 ; +150 (kJ) ;	Allow 150 with 'endothermic' stated. Positive sign needed for the second mark.	2	AO2 4.5.1.2
06.6	no, the reactants and products have the same energy (with or without a catalyst)		1	AO3 4.5.1.2
06.7	provides an alternative pathway / route with a lower activation energy ; more particles have enough energy to react so more successful collisions ;		2	AO2 4.6.1.4
07.1	436 + 242 ; 678 (kJ/mol) ;		2	AO2 4.5.1.3
07.2	2 × 432 ; 864 (kJ/mol) ;		2	AO2 4.5.1.3
07.3	678 − 864 ; −186 (kJ/mol) ;	Allow ecf from Q07.1 and Q07.2. Sign is needed for the second mark.	2	AO2 4.5.1.3
07.4	exothermic ; more energy is released making bonds than is needed to break bonds / the energy change is negative ;		2	AO2 4.5.1.3
08.1	bonds broken: 4 × 413 + 2 × 498 = 2648 ; bonds made: 2 × 805 + 4 × 464 = 3466 ; 2648 − 3466 ; −818 (kJ/mol) ;	Energy change = energy to break bonds − energy released making bonds. Correct answer with no working scores 4. Allow ecf.	4	AO2 4.5.1.3
08.2	945 + 3 × 436 = 2253 (broken) ; 6 × 391 = 2346 (made) ; −93 (kJ/mol) ;	Each ammonia molecule has three N–H bonds, so $2NH_3$ has six.	3	AO2 4.5.1.3
08.3	436 + 193 = 629 (broken) ; 629 − (2 × H–Br) = −103 so 2 × H–Br = 732 ; 366 (kJ/mol) ;	Check: 2 × 366 = 732, and 629 − 732 = −103.	3	AO3 4.5.1.3
08.4	more energy is needed to break bonds than is released making bonds ; so the reaction is endothermic ;		2	AO2 4.5.1.3
09.1	electrolyte		1	AO1 4.5.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
09.2	magnesium ; it has the greatest difference in reactivity from copper ;		2	AO3 4.5.2.1
09.3	0.32 V (1.10 − 0.78) ;	Allow 0.3.	1	AO3 4.5.2.1
09.4	0 V / no voltage ; there is no difference in reactivity between the metals ;		2	AO3 4.5.2.1
09.5	the chemical reaction uses up the reactants ; so the reaction stops / the voltage falls to zero ;	Alkaline batteries are non-rechargeable.	2	AO2 4.5.2.1
09.6	an external electric current is supplied ; this reverses the chemical reactions (regenerating the reactants) ;		2	AO1 4.5.2.1
10.1	water		1	AO1 4.5.2.2
10.2	$2H_2 + O_2 \rightarrow 2H_2O$; correct formulae and balanced ;		2	AO1 4.5.2.2
10.3	P ; hydrogen is supplied at P and loses electrons (to form H^+ ions) ;		2	AO2 4.5.2.2
10.4	P: $2H_2 \rightarrow 4H^+ + 4e^-$; Q: $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$;	Allow $H_2 \rightarrow 2H^+ + 2e^-$ and $½O_2 + 2H^+ + 2e^- \rightarrow H_2O$.	2	AO2 4.5.2.2
10.5	it does not need to be recharged (works while fuel is supplied) ; the only product is water (no pollutants) ;	Allow 'lighter', 'gives a constant voltage', 'less toxic waste'. Any two.	2	AO3 4.5.2.2
10.6	hydrogen is flammable / explosive / difficult to store ; hydrogen may be made using fossil fuels ; expensive ;	Any one.	1	AO3 4.5.2.2
11.1	−9 °C / a fall of 9 °C		1	AO2 4.5.1.1
11.2	carbon dioxide ; turns limewater cloudy / milky ;		2	AO1 4.3.2
11.3	endothermic ; energy is taken in from the surroundings so the temperature falls ;		2	AO2 4.5.1.1
11.4	instant cold packs / cooling packs for injuries		1	AO3 4.5.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • in any reaction energy is needed to break bonds in the reactants (activation energy) • energy is released when new bonds form in the products • exothermic: more energy is released by bond making than is needed to break bonds • so the products are at a lower energy than the reactants and energy is transferred to the surroundings (temperature rises) • endothermic: more energy is needed to break bonds than is released making bonds • so the products are at a higher energy than the reactants and energy is taken in from the surroundings (temperature falls); reaction profile peak shows the activation energy for both		6	AO1 4.5.1.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • fuel cell: only product is water so no carbon dioxide or pollutants at the point of use • fuel cell: can be refuelled with hydrogen quickly; does not need long charging times; keeps a constant voltage • fuel cell: hydrogen is flammable and hard to store (high pressure); few filling stations; may be made from fossil fuels • battery: can be recharged from the mains / renewable electricity; no need to store a flammable gas • battery: must be recharged (takes time), loses capacity with age, heavy; contains toxic metals that must be recycled • a justified conclusion, e.g. fuel cells better for long distances / quick refuelling; batteries better where charging points exist		6	AO3 4.5.2.2

Where marks are lost in Topic 4.5

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Exothermic / endothermic:** say where the energy goes. Exothermic: energy is transferred *to* the surroundings and the temperature rises. Endothermic: energy is taken in *from* the surroundings and the temperature falls.
- ☐ **Reaction profiles:** activation energy is from the reactants to the *top of the peak*. Overall energy change is from the reactants level to the products level. Draw arrows with arrowheads and label them.
- ☐ **Drawing profiles:** endothermic means the products line is *higher* than the reactants. Always draw the peak above both lines.
- ☐ **Bond energy sums:** energy change = (energy to break bonds) − (energy released making bonds). Write both totals, then subtract. A negative answer means exothermic. Count every bond in every molecule ($2NH_3$ has six N–H bonds).
- ☐ **Catalysts:** they lower the activation energy by giving an alternative pathway. They do *not* change the overall energy change.
- ☐ **Required practical:** insulate (polystyrene cup + lid), stir, record the starting temperature and the highest temperature, and use the temperature *change*. Repeat to get a mean.
- ☐ **Cells:** the bigger the difference in reactivity between the two metals, the bigger the voltage. Batteries stop because the reactants are used up.
- ☐ **Fuel cells:** hydrogen is oxidised at the negative electrode; oxygen is reduced at the positive. The only product is water. Know one advantage *and* one disadvantage.
- ☐ **Signs and units:** include + or − in answers for ΔH and use kJ/mol. Graph questions: use a ruler, draw lines of best fit that meet, and read the values to half a square.